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EXP NO : 4 **QUERYING LEDGER DATA WITH HYPERLEDGER FABRIC**

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Aim:

To query and retrieve ledger data from a Hyperledger Fabric blockchain network using JavaScript chaincode, and verify asset information stored in the world state database.

Requirements

- Kali Linux / Ubuntu
- Docker and Docker Compose
- Node.js (Version 18 or later) and npm
- Git
- Hyperledger Fabric Samples, Binaries, and Docker Images

Procedure & Execution

1. **Navigate to the Test Network** Command(s):

```
┌──(student㉿linux)-[~]
└─$ cd ~/fabric-dev/fabric-samples/test-network
```

2. **Start the Hyperledger Fabric Network** Command(s):

```
┌──(student㉿linux)-[~]
└─$ ./network.sh up createChannel -ca

Creating channel 'mychannel'
Starting peer0.org1
Starting peer0.org2
Starting orderer.example.com
Network Started Successfully
```

3. **Deploy the Asset Transfer Chaincode** Command(s):

```
┌──(student㉿linux)-[~]
└─$ ./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript

Packaging Chaincode
Installing Chaincode
Approving Chaincode
Committing Chaincode
Chaincode committed successfully
```

4. **Initialize the Ledger** Command(s):

```
┌──(student㉿linux)-[~]
└─$
```

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```
┌──(student㉿linux)-[ ]
```

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```
└─$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --
```

Ledger Initialized Successfully

5. **Query All Assets**
Command(s):

```
┌──(student㉿linux)-[~]
└─$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[
  {
    "ID": "asset1",
    "Color": "blue",
    "Size": 5,
    "Owner": "Tomoko",
    "AppraisedValue": 300
  },
  {
    "ID": "asset2",
    "Color": "red",
    "Size": 5,
    "Owner": "Brad",
    "AppraisedValue": 400
  }
]
```

6. **Query a Specific Asset**
Command(s):

```
┌──(student㉿linux)-[~]
└─$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset", "asset1"]}'
{
  "ID": "asset1",
  "Color": "blue",
  "Size": 5,
  "Owner": "Tomoko",
  "AppraisedValue": 300
}
```

7. **Create a New Asset**
Command(s):

```
┌──(student㉿linux)-[~]
└─$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --
cafile $ORDERER_CA -C mychannel -n basic -c '{"function":"CreateAsset", "Args":["asset20",
"Green","8","User","1500"]}'
Asset asset20 created successfully
```

8. **Query the Newly Created**
Asset Command(s):

```
┌──(student㉿linux)-[~]
└─$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
List of all assets including asset20 displayed successfully.
```

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```
┌──(student㉿linux)-[~]
└─$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset20"]}'
{
  "ID": "asset20",
  "Color": "Green",
  "Size": 8,
  "Owner": "User",
  "AppraisedValue": 1500
}
```

9. Query All Assets Again

Command(s):

10. Query a Non-Existing Asset

Command(s):

```
┌──(student㉿linux)-[~]
└─$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset50"]}'
Error: The asset asset50 does not exist.
```

11. Stop the Fabric Network

Command(s):

```
┌──(student㉿linux)-[~]
└─$ ./network.sh down
Removing containers
Removing Docker Network
Network stopped successfully
```

Result:

The ledger data stored in the Hyperledger Fabric blockchain network was successfully queried using JavaScript chaincode. Asset records were retrieved from the world state database, specific asset details were verified, and newly created assets were successfully queried, demonstrating efficient data retrieval and verification in a permissioned blockchain environment.